

# Global CO<sub>2</sub> Emissions: Trends, Regional Shifts, and Per Capita Inequalities

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A data-driven analysis of fossil fuel CO<sub>2</sub> from the Industrial Revolution to 2024

Scope: 1750 – 2024

Data sources: Global Carbon Budget 2025 | NASA Mauna Loa Observatory | Our World in Data

# Atmospheric CO<sub>2</sub> Reaches 429 ppm — 50% Above Pre-Industrial Levels

# 429

ppm CO<sub>2</sub> (Feb 2026, Mauna Loa)

+53% above pre-industrial baseline of ~280 ppm

## Timeline Context

**Pre-industrial (~1750):** ~280 ppm

**1958 – Keeling Curve begins:** 315 ppm

**2000:** 369 ppm

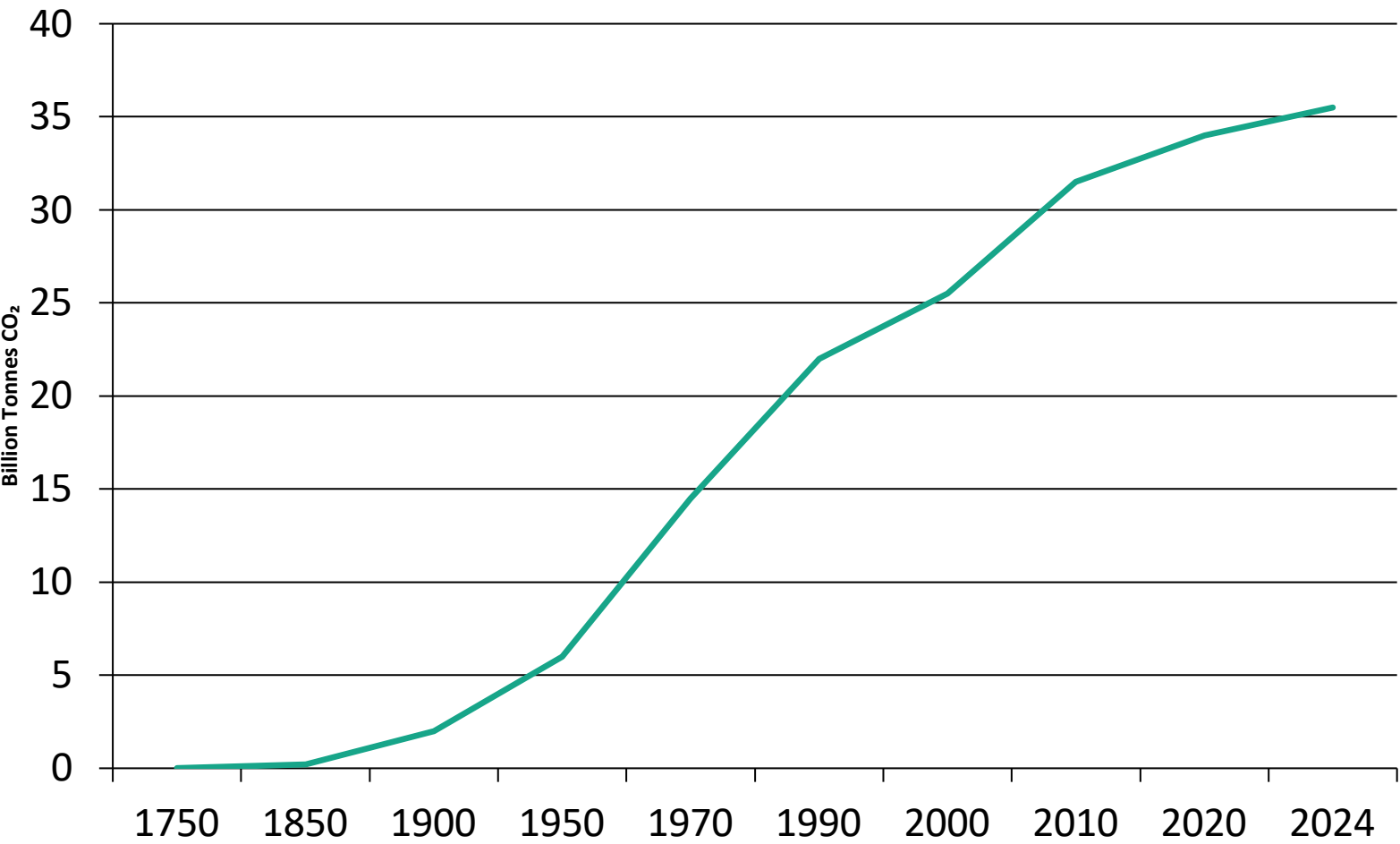
**2015 – Paris Agreement:** 400 ppm crossed

**2026 (latest):** 429 ppm ▲

■ The rise from 280 to 429 ppm occurred in ~200 years — a geological blink. The same change at the end of the last ice age took thousands of years.

# Global Fossil Fuel CO<sub>2</sub> Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Global CO<sub>2</sub> (Billion Tonnes)



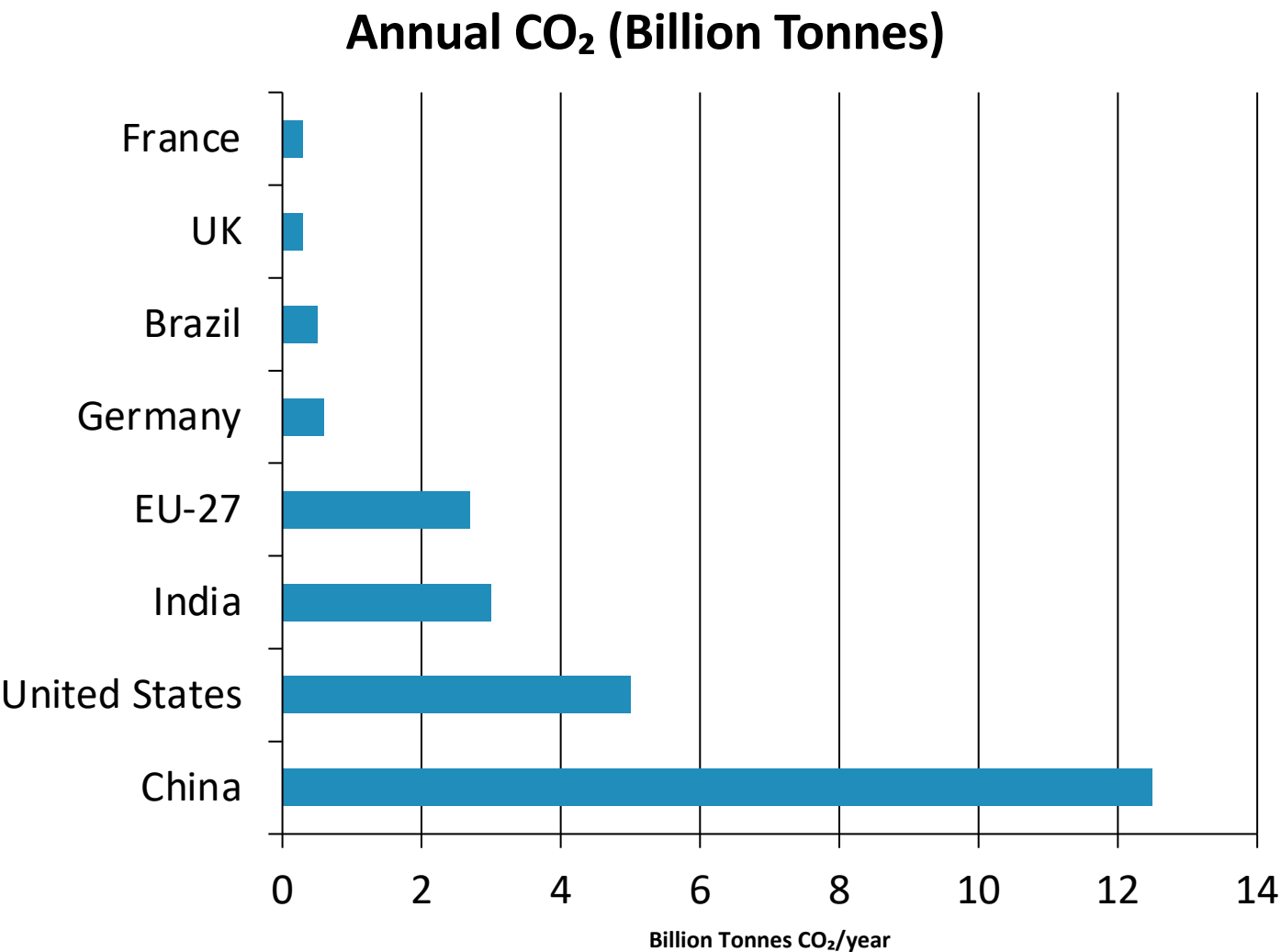
1950: 6 Bt

1990: 20+ Bt

2024: 35+ Bt

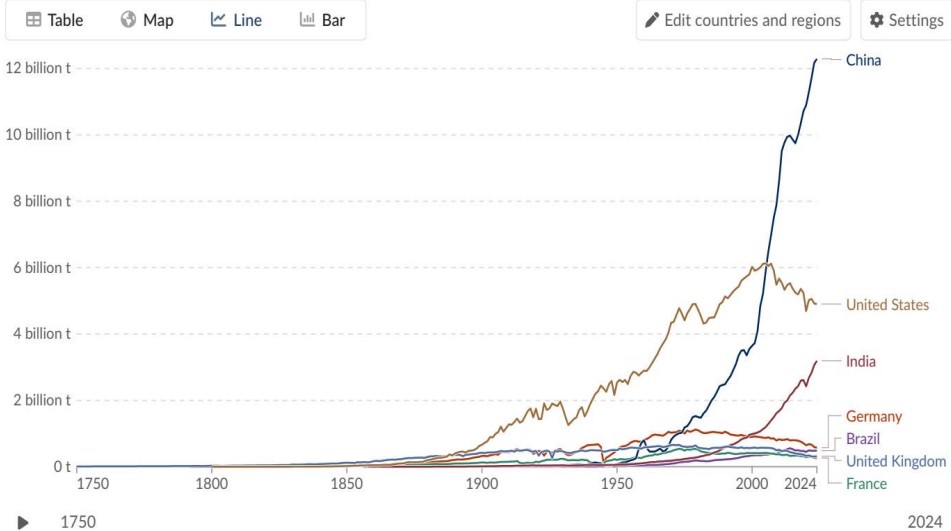
Emissions have not yet peaked despite slowing growth rates. A sustained decline requires structural transformation of the global energy system.

# China Now Emits Over 12 Billion Tonnes — More Than Double the United States



### Annual CO<sub>2</sub> emissions

Carbon dioxide (CO<sub>2</sub>) emissions from fossil fuels and industry. Land-use change emissions are not included.



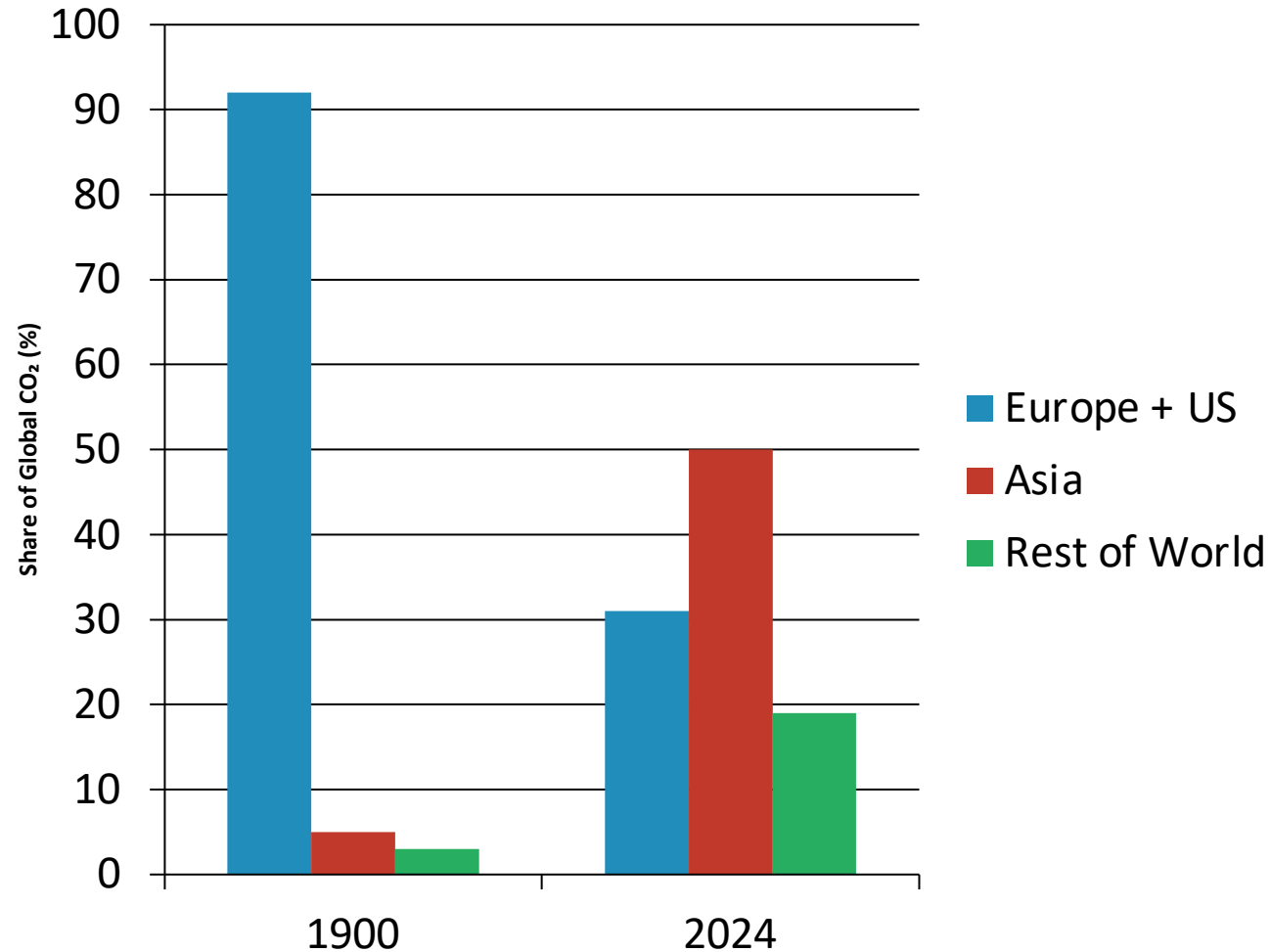
Data source: Global Carbon Budget (2025) — [Learn more about this data](#)  
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Figure 3 — Annual CO<sub>2</sub> by country (Our World in Data)

**China:** ~27% of global total; still rising  
**United States:** peaked ~2005 at 6 Bt; now ~5 Bt  
**India:** ~3 Bt; trajectory steeply upward  
**EU-27:** ~2.7 Bt; declining trend

# Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



## A Century of Seismic Shift

### 1900:

Europe + US held >90% of global emissions  
Asia contributed only ~5%

### 2024:

Asia produces ~50% of global CO<sub>2</sub>  
US + Europe combined: <33%  
Africa + S. America each: ~3–4%

### Why it matters:

Emissions responsibility has shifted, but historical cumulative contributions still weigh heavily on industrialised nations.

# Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

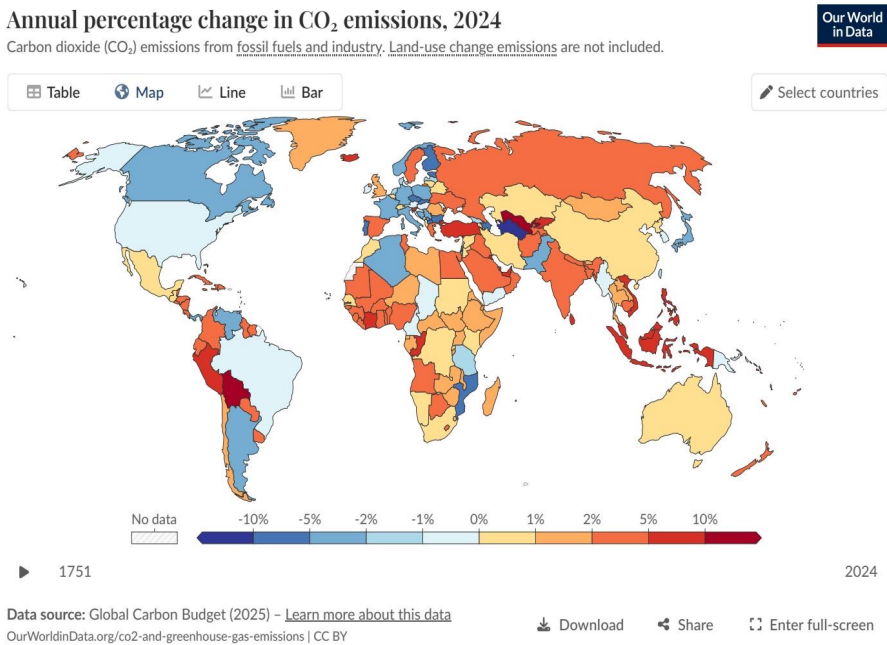
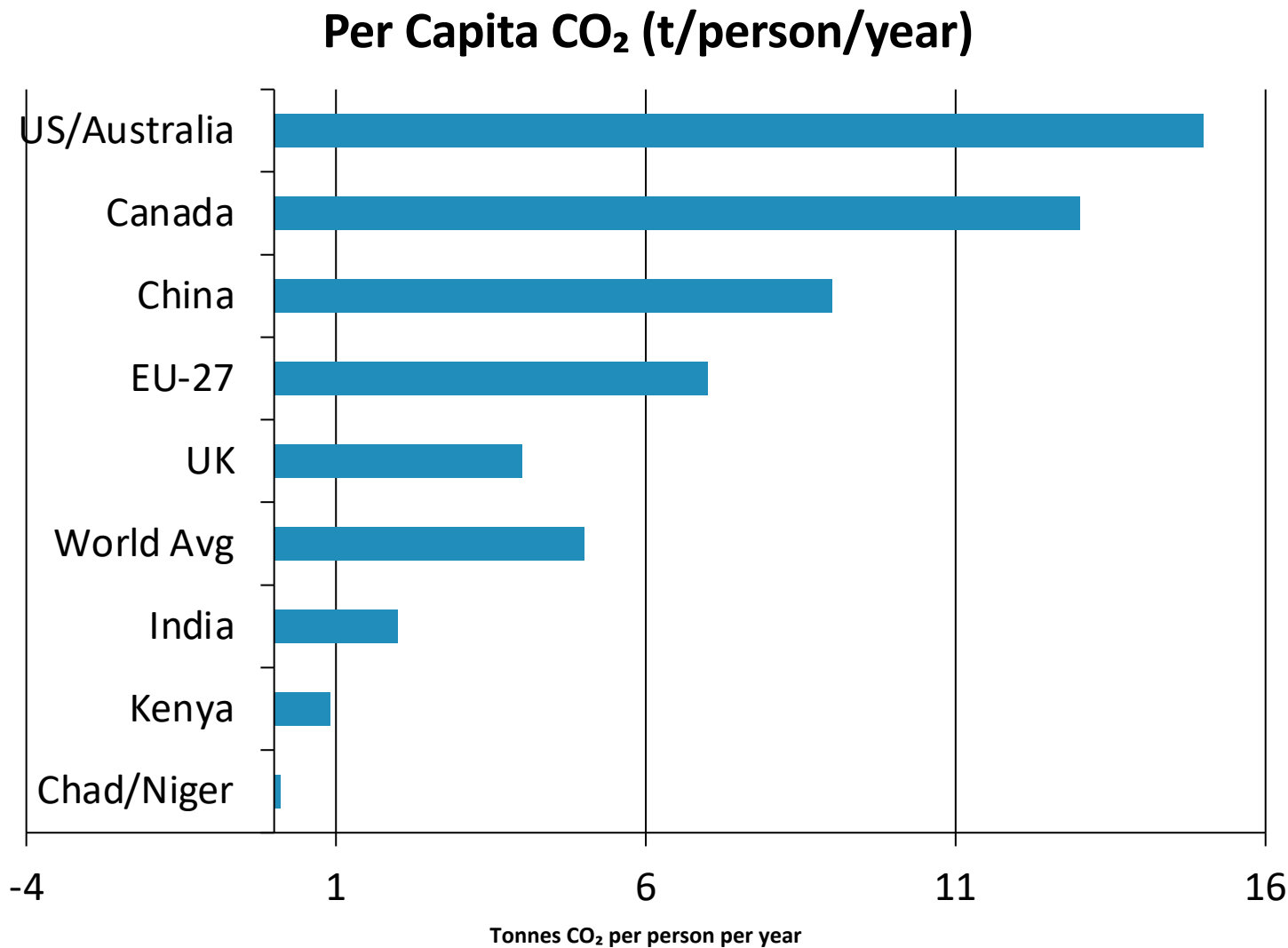
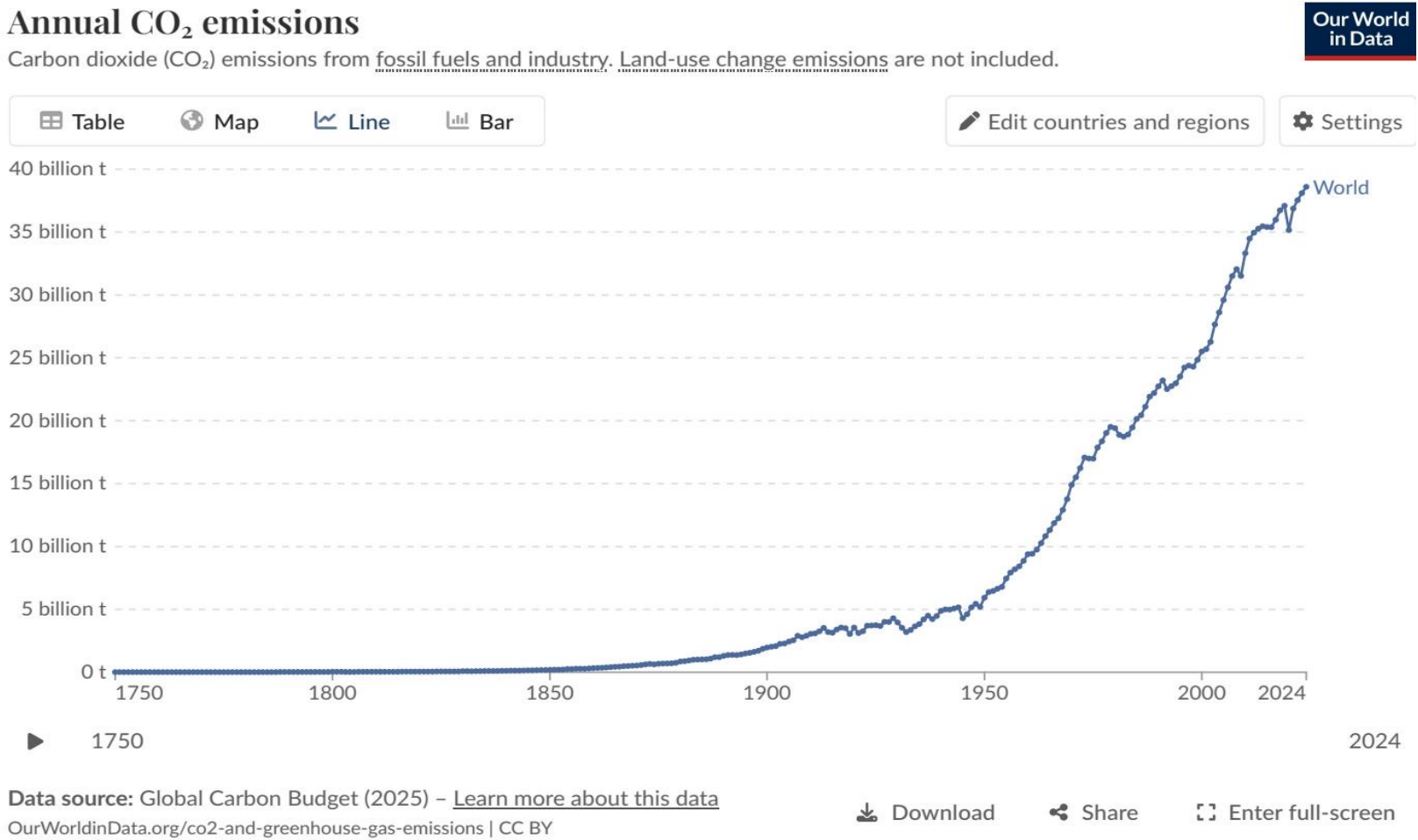


Figure 1 — Per capita CO<sub>2</sub> over time (Our World in Data)

The average American emits in under two days what the average person in Chad emits in an entire year.  
150× gap between top and bottom emitters.

# 2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



## Map Legend

**Blue (–1% to –5%)**

US, most of Europe, UK

**Orange/Red (+2% to +10%+)**

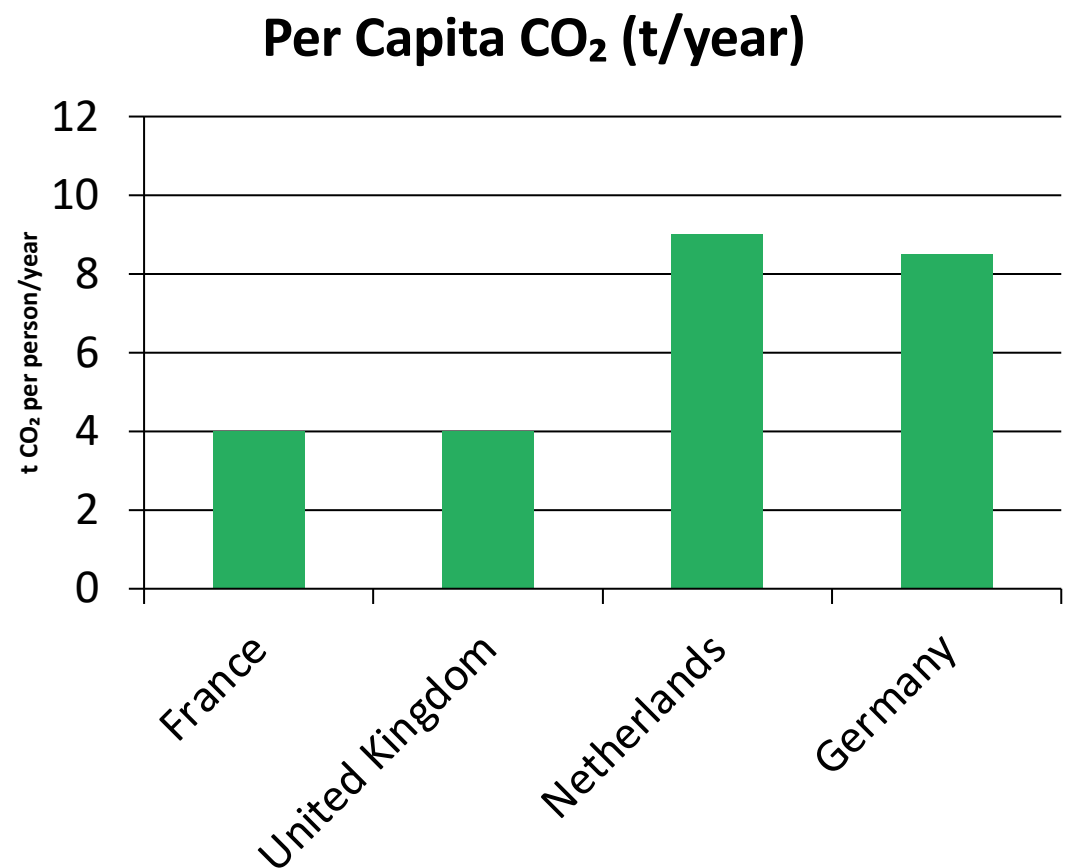
Russia, SE Asia, parts of Africa

### Key findings:

- US: –1% to –3% decline
- EU: broad declines across members
- Russia: continued increases
- SE Asia: +2% to +10% growth
- Some S. America nations: >10% spike
- Africa: mixed picture

Figure 4 — Annual % change in CO<sub>2</sub> emissions, 2024 (Our World in Data)

# Energy Mix Explains Why France Emits Far Less Per Capita Than Germany



Green bars = high nuclear/renewables share  
Orange bars = higher fossil dependence

| Country     | Nuclear / Renewables share | Fossil fuel share | Per Capita CO <sub>2</sub> |
|-------------|----------------------------|-------------------|----------------------------|
| France      | ~85% (75% nuclear)         | ~15%              | ~4 t                       |
| UK          | ~50%+ (nuclear+wind)       | ~50%              | ~4 t                       |
| Germany     | ~50% (rising renew.)       | ~50%              | ~8.5 t                     |
| Netherlands | ~35%                       | ~65%              | ~9 t                       |

Similar GDP per capita, very different CO<sub>2</sub> outcomes — policy and energy choice matter.



# Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

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1

Global emissions now exceed 35 billion tonnes per year and have not yet peaked, despite recent slowing in growth rates.

2

China alone accounts for >25% of global CO<sub>2</sub> (~12.5 Bt) — more than double the United States (~5 Bt). India (~3 Bt) is third and rising fast.

3

The geographic centre of emissions has shifted dramatically: from >90% in Europe + US in 1900 to Asia producing ~50% today.

4

Per capita inequalities remain extreme — a 150× gap separates the US/Australia (~15 t/person) from Chad/Niger (~0.1 t). Equity must inform negotiations.

5

Policy and energy mix — not just prosperity — determine per capita outcomes. France and Germany have similar GDPs but France emits roughly half as much per capita.